UNIVERSITÉ DES SCIENCES ET DE LA TECHNOLOGIE HOUARI BOUMEDIENE

USTHB / FACULTÉ DE GÉNIE ÉLECTRIQUE / DÉPARTEMENT AUTOMATIQUE

ROBOTIQUE ET SYSTÈMES INTELLIGENTS (RSI) / FOURTH-YEAR ENGINEERING

IntelQCopter

Autonomous Quadcopter

PERSONAL AND PROFESSIONAL PROJECT / RSI7.8



AI CONCEPT ILLUSTRATION / RED-WHITE F450 / NOT A TEST PHOTOGRAPH

RP2350	PID	ROS 2	Gazebo
FLIGHT CONTROLLER	STABILITY CONTROL	ONBOARD NAVIGATION	VALIDATION TARGET

RP2040 handheld / LoRa primary link / reserved CC1101 channels

Companion computer sought: Raspberry Pi 4/5 or NVIDIA Jetson + camera.

Stable manual operation remains independent of the companion computer.

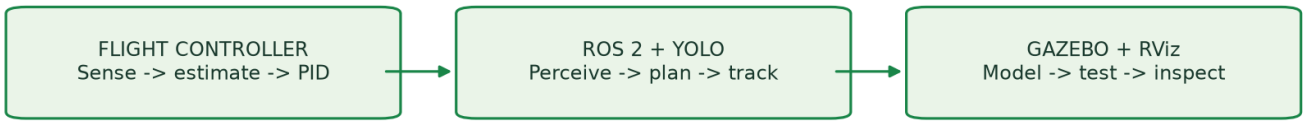
27 OCTOBER 2026 / ACADEMIC YEAR 2026-2027 / EDITION 6.0

Student: _____ Supervisor: _____

Proposed design and measured results are labelled. Flight acceptance remains pending.

github.com/BenMohStem/IntelQCopter / university report

IntelQCopter in one page



PROPOSED ARCHITECTURE / Three complementary tasks: the FC stabilizes, the companion navigates, the simulator evaluates.

Purpose: build a quadcopter that flies stably under pilot commands, then navigates between approved waypoints while avoiding observed obstacles. **Learning:** apply feedback control, identification, estimation and experimental validation to one integrated engineering project.

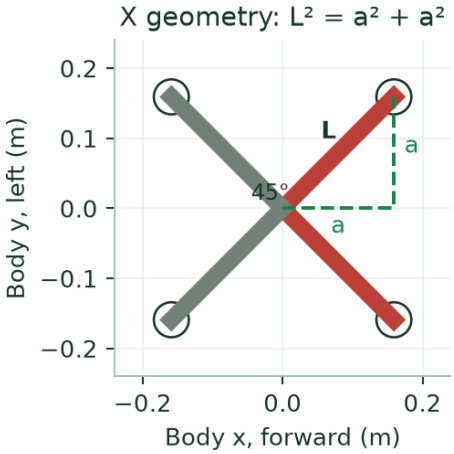
Reading route	Essential question	Pages
1. Flight controller	How does the aircraft sense, estimate and stabilize?	3-10
2. ROS 2 and YOLO	How does the onboard computer perceive and navigate?	11-13
3. Gazebo and RViz	How do the model and actual software get tested?	14-15
4. Resources	What hardware, software and development method are needed?	16-17
5. Handheld and communication	How do pilot and companion commands reach the FC?	18-19

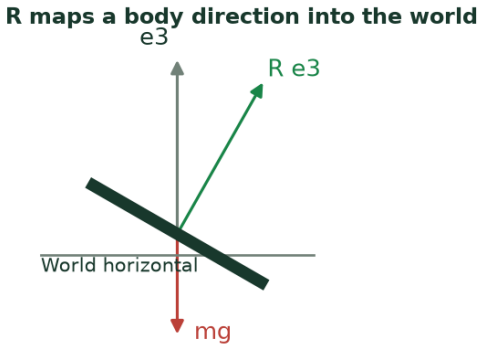
Target: RP2350 aircraft; RP2040 handheld. Stable manual flight works without the companion. Raspberry Pi 4/5 or NVIDIA Jetson is being sought for autonomous operation.

Evidence: secured sensor recordings and offline analyses exist; Pico integration and autonomous flight remain unvalidated. Figures distinguish recorded evidence, analytic examples and proposed designs. Physical motor outputs remain disabled.

Notation: bold = vector/matrix; dot = time derivative; hat = fitted estimate; bar = sample mean; T superscript = transpose; -1 superscript = inverse; || || = vector length. Symbols are defined locally; the same letter can have a different role on another page.

1. Flight controller | F450 physics and geometry





ANALYTIC GEOMETRY / L is centre-to-motor distance; a is its horizontal projection. R redirects the body thrust axis into world coordinates.

$$a = L\cos 45^\circ = \frac{L}{\sqrt{2}} = 0.1591 \text{ m}, \quad L = 0.225 \text{ m}$$

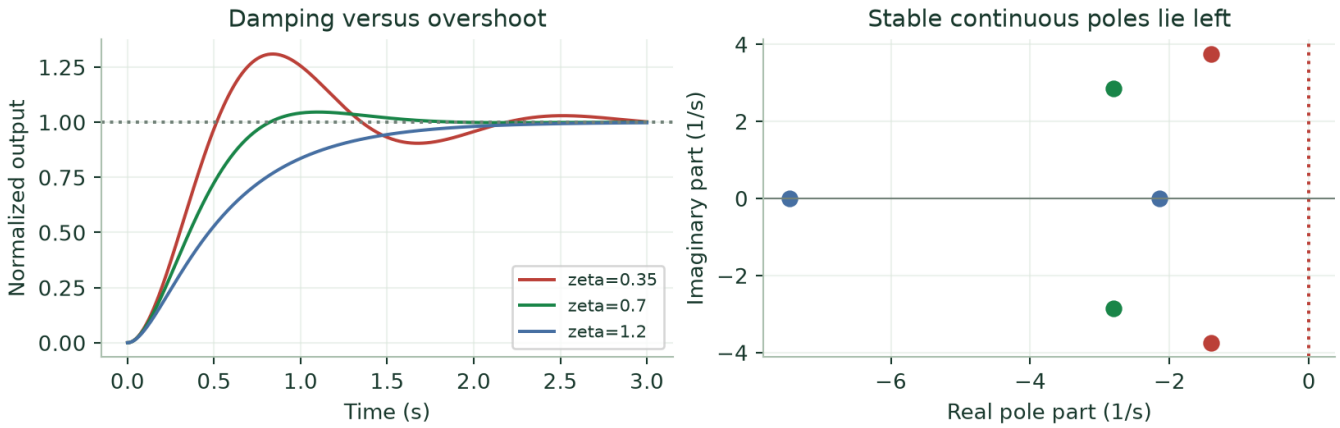
$$m\dot{\mathbf{v}} = R\mathbf{e}_3 T - mg\mathbf{e}_3 - D\mathbf{v} + \mathbf{f}_{ext}$$

R rotates directions: R is a dimensionless 3 x 3 rotation matrix; thrust points along the drone's upward axis; R expresses that axis in the world frame. At level hover, R e3 = e3 and T = mg. **D damps motion:** D = diag(dx,dy,dz) is a local linear drag approximation, not a measured constant. Drag is -D v; each d has units N s/m.

L is an arm length, not the frame diagonal: nominal 450 mm opposite-motor spacing gives L = 225 mm from centre to motor. On an X frame, the two horizontal projections are equal: L² = a² + a², hence a = L/sqrt(2). Coordinates (±a,±a) determine torque lever arms.

Symbols: m: mass (kg); v: world velocity (m/s); T: total thrust (N); e3 = [0,0,1] transpose; g: gravity (9.80665 m/s² assumed); f_ext: disturbance force (N). World: ENU; body: FLU; mounting and signs require physical checks. Here R is rotation, not sensor covariance.

1.1 Physics | torque and PID design model



ILLUSTRATIVE LTI MODEL / Educational second-order responses, not tuned drone gains or simulated aircraft trials.

$$T = \sum_{i=1}^{4} T_i, \quad \boldsymbol{\tau}_i = \mathbf{r}_i \times \mathbf{F}_i, \quad J\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (J\boldsymbol{\omega})$$

$$J_{xx}\ddot{\phi} \simeq \tau_x, \quad J_{yy}\ddot{\theta} \simeq \tau_y, \quad J_{zz}\dot{r} \simeq \tau_z, \quad m\ddot{z} \simeq \Delta T$$

$$G_\phi(s) = \frac{1}{J_{xx}s^2}, \quad G_\theta(s) = \frac{1}{J_{yy}s^2}, \quad G_r(s) = \frac{1}{J_{zz}s}, \quad G_z(s) = \frac{1}{ms^2}$$

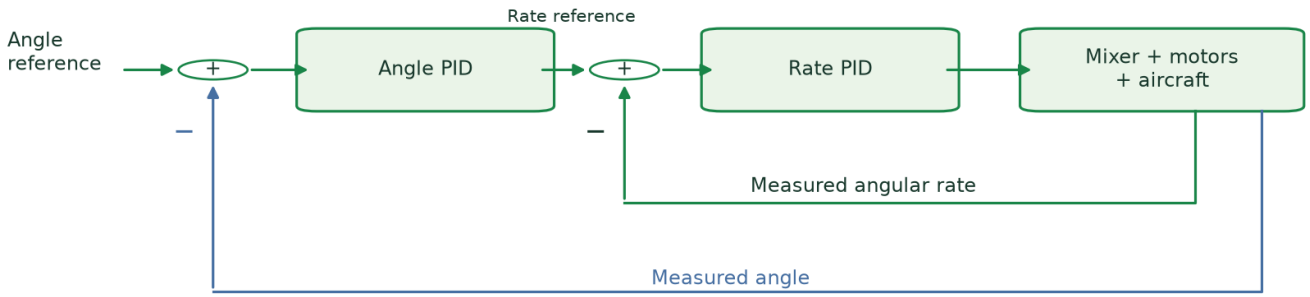
Supplied simulation inputs	Value	Interpretation
m; stated mass standard deviation	1.038 kg; 0.0160312 kg	Not independently measured
Jxx = Jyy; Jzz	0.006421; 0.011853 kg m²	Approximate COM inertias
Hover total / per rotor	10.179 / 2.545 N	mg, assuming equal loading

Symbols: i: rotor; r_i: lever-arm vector (m); F_i: rotor force (N); τ: torque (N m); J: inertia matrix (kg m²); ω: angular velocity (rad/s); φ/θ: roll/pitch (rad); r: yaw rate (rad/s); z: height (m); ΔT = T - mg; s: Laplace variable (1/s); G: input-to-output transfer function. Rotor reaction torque supplies yaw.

Gain example: for one ideal torque-controlled axis with PD feedback, J here is the selected axis inertia; matching the closed-loop denominator gives Kp = J ωn² and Kd = 2 ζ ωn J. Here Kp is in N m/rad and Kd in N m s/rad. These are design relations, not identified flight gains.

Design scope: zero-initial-condition Laplace transform gives Jxx s² Φ = Tx, hence Gφ = Φ/Tx. Φ and Tx are the Laplace transforms of φ and τx. Hover linearization assumes small angles/rates and principal axes; it is not a global model. Identify actuator lag Gact = 1/(τm s + 1), where τm is motor/ESC time constant (s), plus delays, thrust curve and saturation. Final gains remain pending those measurements. Plot: educational damping example; ζ is damping ratio, ωn natural frequency.

1.2 Control | cascaded PID and digital execution



PROPOSED CASCADED FEEDBACK / Roll/pitch each use an outer angle and inner rate loop. Yaw uses rate feedback; height supplies the vertical-speed loop.

$$e_k = r_k - y_k, \quad u_k = K_p e_k + I_k + K_d \frac{e_k - e_{k-1}}{\Delta t}, \quad I_{k+1} = I_k + K_i e_k \Delta t$$

$$u = \mathrm{clip}(u_{raw}, u_{min}, u_{max}), \quad I_{k+1} \leftarrow I_{k+1} + K_{aw}(u - u_{raw})\Delta t$$

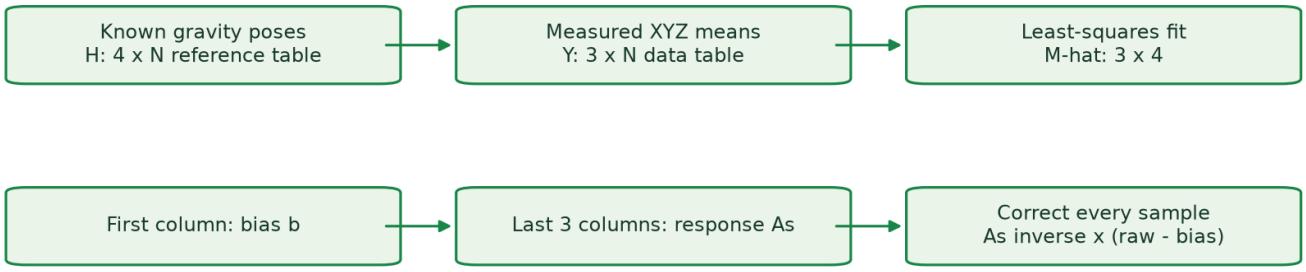
Command	Feedback law	Centred-stick behaviour
Roll / pitch	Angle PID -> angular-rate PID	Level attitude; no horizontal brake/position hold
Yaw	Angular-rate PID	Zero commanded yaw rate
Vertical	Height reference -> vertical-speed PID	Maintain height through vertical loop
RT / LT	Bounded increase / decrease	Gentle profile; retain correction authority

Why PID: I chose it to apply learned error feedback, stability, integral error rejection and derivative damping. Six PID blocks: 2 angle + 2 rate + 1 yaw-rate + 1 vertical-speed; the outer height-to-speed law remains to finalize.

Symbols: k: sample index; Δt: actual interval (s); r/y/e: reference/output/error; u: requested actuation; I: integral contribution; Kp/Ki/Kd: loop gains; Kaw: anti-windup gain (1/s); clip: enforce bounds. Filter the derivative in implementation; the difference above explains its unfiltered principle.

1 kHz target: timestamp acquisition -> estimate -> PID -> mixer. Proposed collective ceiling 1700-1800 µs reserves PID headroom; it is not measured thrust. Proposed margins: phase >=45°, gain >=6 dB; overshoot <=10%; settling within 5% <=1 s. Numerical gains require identification and closed-loop tests.

1.3 Sensors | six-pose calibration explained



CALIBRATION METHOD / Known reference poses and measured XYZ means identify a bias vector and response matrix. Hold out a repeat to evaluate residuals.

$$\mathbf{r}_j = \mathbf{b} + A_s\mathbf{a}_j + \boldsymbol{\varepsilon}_j, \quad \widehat{M} = [\widehat{\mathbf{b}}\ \widehat{A}_s], \quad \mathbf{a}_{cor} = \widehat{A}_s^{-1}(\mathbf{r} - \widehat{\mathbf{b}})$$

$$\mathbf{h}_j = [1, a_{x,j}, a_{y,j}, a_{z,j}]^T, \quad H = [\mathbf{h}_1 \cdots \mathbf{h}_N], \quad Y = [\mathbf{r}_1 \cdots \mathbf{r}_N]$$

$$Y \simeq MH, \quad \widehat{M} = YH^T(HH^T)^{-1}$$

What are H, Y and M-hat? H stores the known reference poses, one column per pose; Y stores the corresponding measured three-axis means. M-hat is the fitted 3 x 4 coefficient table: its first column is bias b; its last three columns are sensor response As. The hat means estimated, not exact. For N fit poses: H is 4 x N; Y is 3 x N.

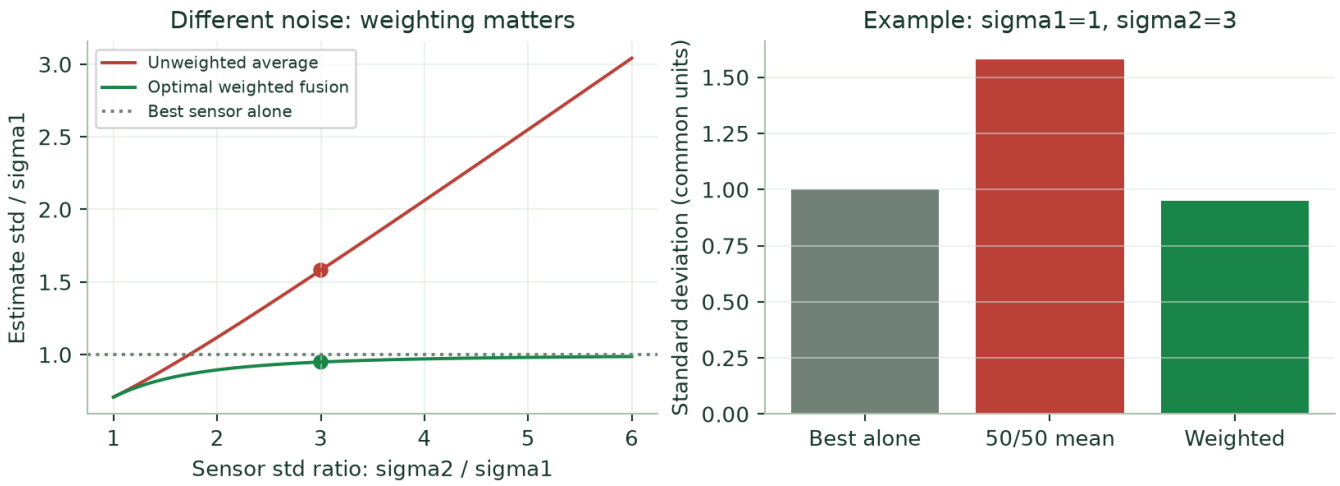
Why this fit? Choose M to minimize the sum of squared residuals ||Y - MH||². Differentiating and setting the gradient to zero gives M H H-transpose = Y H-transpose. Full row rank of H allows inversion, producing the formula. Then invert As to correct all three axes together.

Symbols: j: pose; r_j: raw acceleration; a_j: known gravity reference (m/s²); ε_j: residual; b: bias (m/s²); As: dimensionless response; a_cor: corrected acceleration. Six orientations ±g along X/Y/Z reveal offsets, scales and cross-axis effects. One pose cannot.

Sensor roles: MPU6050 measures acceleration/rate; BMP280 supplies pressure-derived altitude; GY-271 supplies a magnetic heading observation after chip identification/calibration; lidar supplies one range ray.

Evidence [P1]: 18 x 20 s recordings; repetitions 1-2 fit, third held out. Fixture uncertainty remains. MPU registers are volatile: restore/read back at boot; do not subtract bias twice. Full matrix correction stays in software. Gyro needs separate motionless bias plus known signed-rotation checks; those rotation checks remain pending.

1.4 Estimation | why weighted fusion beats averaging



ANALYTIC INDEPENDENT-SENSOR EXAMPLE / Inverse-variance fusion improves the model variance. Correlation, bias, different units or wrong dynamics invalidate the simple comparison.

For two unbiased independent measurements of the same scalar x, let y1 = x + e1 and y2 = x + e2, with variances R1 and R2. Choose an unbiased linear estimate xhat = (1-w)y1 + w y2; its variance is minimized by inverse-variance weighting.

$$P(w) = (1-w)^2 R_1 + w^2 R_2, \quad \frac{dP}{dw} = 0 \implies w^* = \frac{R_1}{R_1 + R_2}$$

$$P^* = \frac{R_1 R_2}{R_1 + R_2} \leq \min(R_1, R_2), \qquad P_{average} = \frac{R_1 + R_2}{4}$$

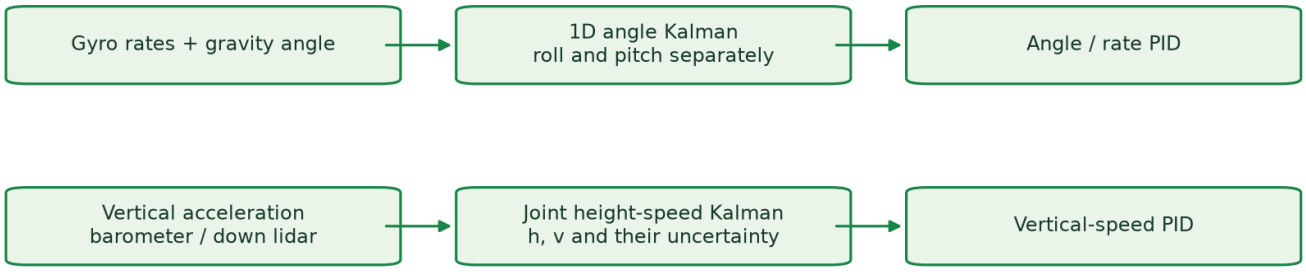
$$\mathbf{w}^* = \frac{R^{-1}\mathbf{1}}{\mathbf{1}^T R^{-1}\mathbf{1}}, \qquad P^* = (\mathbf{1}^T R^{-1}\mathbf{1})^{-1} \quad \text{(full covariance)}$$

Symbols: x: true scalar; y1/y2: observations; e1/e2: errors; R1/R2: error variances (measurement units squared); w: weight of sensor 2; P: estimate-error variance; σ: standard deviation; R in the matrix formula: full observation-error covariance; bold 1: vector of ones; bold w: weight vector; *: minimum-variance choice; P_average: equal-weight variance.

Example: sigma1 = 1 and sigma2 = 3 in the same units. A 50/50 average gives sigma = 1.581, worse than sensor 1 alone. Optimal weighting gives sigma = 0.949. The noisier sensor contributes only 10 percent. No Gaussian assumption is required for this minimum-variance linear weighting under the stated covariance model.

Kalman connection: a prediction is fused with an observation using uncertainty and dynamics. Gyro rate and acceleration cannot be averaged directly: their units and models differ. Integrate the gyro and derive a valid gravity-angle observation first. With equal independent static-sensor variance, the ordinary mean is already optimal among unbiased linear combinations. Known correlation requires the full positive-definite covariance R; bias or model mismatch can invalidate the predicted physical improvement.

1.5 Estimation | attitude and height Kalman filters



PROPOSED ARCHITECTURE / Estimation supplies feedback; PID supplies control. The estimated height and vertical speed feed PID loops.

$$\theta_k^- = \theta_{k-1} + \omega_k \Delta t, \quad K_k = \frac{P_k^-}{P_k^- + R_k}, \quad \theta_k = \theta_k^- + K_k(z_k - \theta_k^-)$$

Attitude: integrate gyro rate to predict angle; correct using an accelerometer-derived gravity angle. P-minus = prior angle-error variance; R = observation variance; K = dimensionless trust. θ/z are estimated/observed angles (rad); ω is gyro rate (rad/s); k is sample; Δt is interval (s). Superscript minus means prediction. Prediction variance adds process uncertainty Q each step.

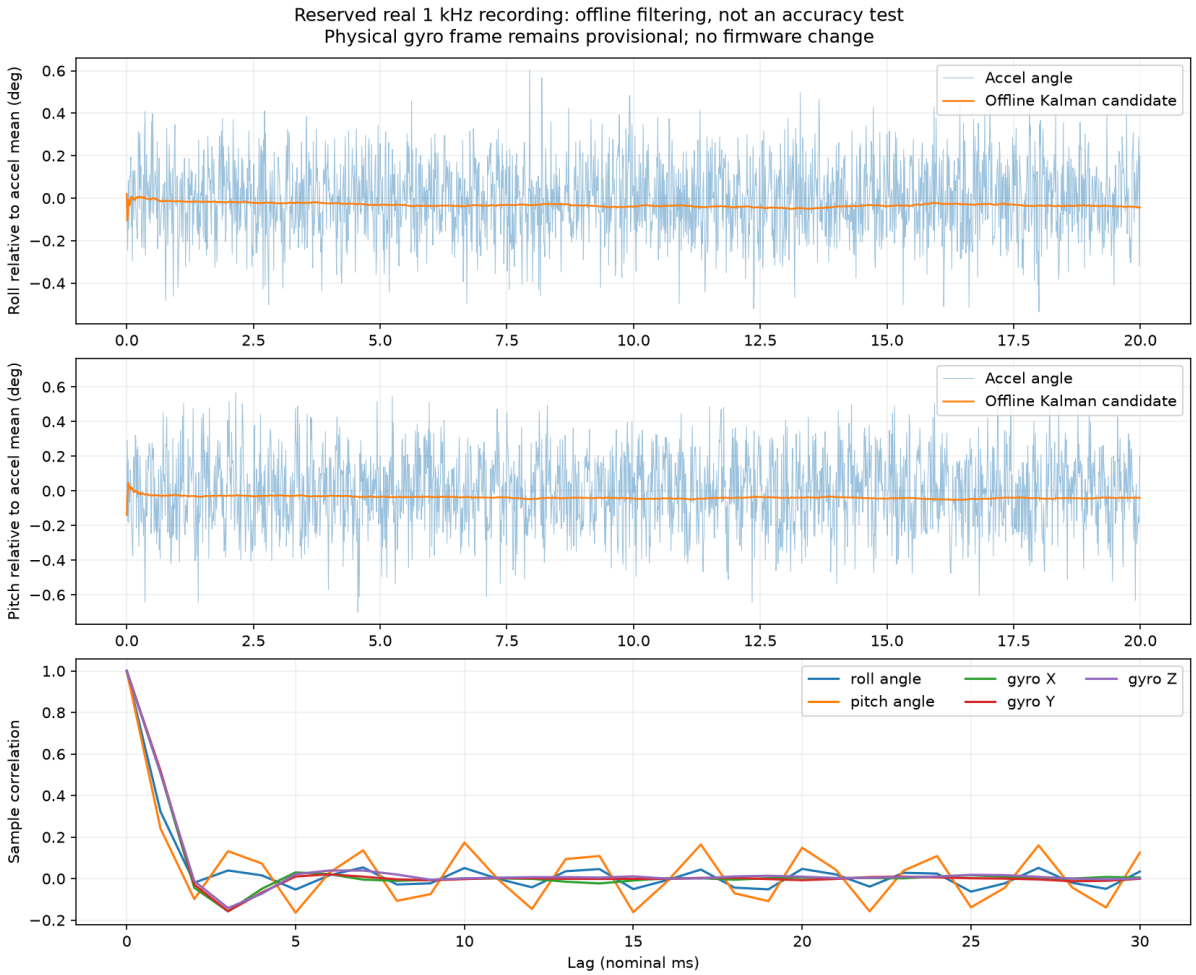
$$h_k^- = h_{k-1} + v_{k-1}\Delta t + \tfrac{1}{2} a_k \Delta t^2, \quad v_k^- = v_{k-1} + a_k \Delta t$$

$$K_h = \frac{P_{hh}^-}{P_{hh}^- + R_h}, \quad K_v = \frac{P_{vh}^-}{P_{hh}^- + R_h}, \quad h_k = h_k^- + K_h \nu_k, \quad v_k = v_k^- + K_v \nu_k$$

Height: v = measured height - predicted height. h/v/a = world height (m), vertical speed (m/s), gravity-compensated acceleration (m/s²). Phh = height-error variance (m²); Pvh = speed/height error covariance (m²/s); Rh = height-observation variance (m²); Kh dimensionless; Kv in 1/s. Propagate both uncertainties and their covariance; this joint height-speed estimate feeds the vertical PID.

Sensor limits: reject invalid/stale observations; barometer drift, lidar tilt/terrain and dropouts need tests. Bottom-versus-front lidar placement remains open. Historical scalar K values [P4], roll 0.00061129/pitch 0.00048326, are conditional stationary-model values, not accepted flight tuning.

1.6 Evidence | real sensor replay and its limits

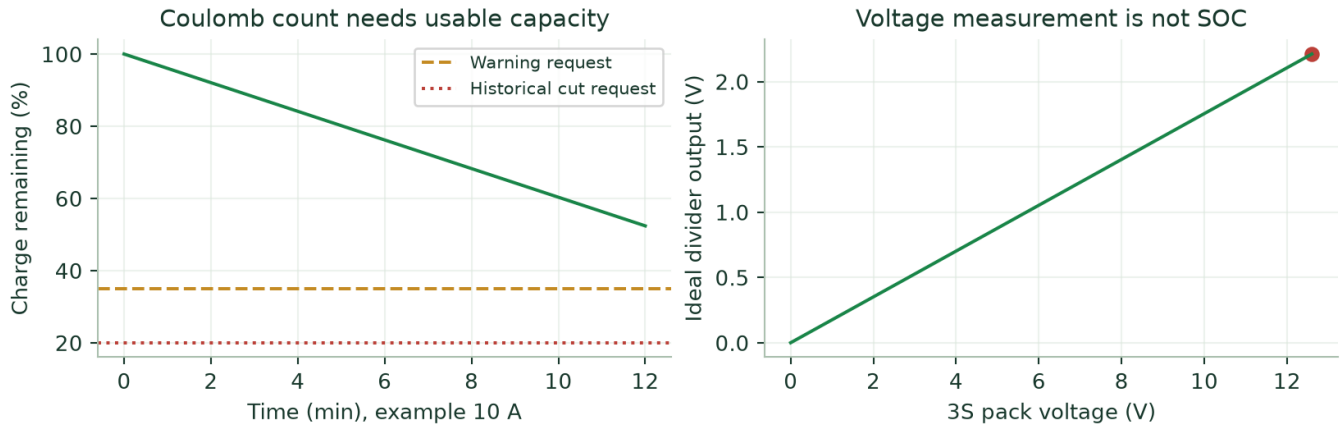


Reserved replay [P4]	Roll	Pitch
Raw angle standard deviation	0.177569 degrees	0.213074 degrees
Filter output standard deviation	0.007000 degrees	0.005692 degrees
Filter minus observation mean	-0.033769 degrees	-0.039908 degrees

Engine: original Python float64 replay of reserved real 1 kHz data; unchanged plot. Not MCU execution or flight. Fixed 2 s initialization excluded. No independent attitude reference: quieter output does not prove accuracy.

Adverse test [P4]: synthetic 0.1g lateral acceleration produced 5.191 degrees peak error; acceleration plus gyro bias produced 5.355 degrees. Apparent gravity tilt is arctan(0.1) = 5.71 degrees while magnitude is only 1.005g. Norm gating alone is insufficient. Dynamic acceptance remains blocked.

1.7 Energy | battery estimation and avionics safety



ANALYTIC EXAMPLE / This is not measured endurance. Whole-pack monitoring cannot protect each cell; a motor cut in flight causes a fall.

$$SOC(t) = SOC(0) - \frac{1}{C_{usable}} \int_0^t I(\tau)\, d\tau, \qquad V_{ADC} = V_{pack} \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

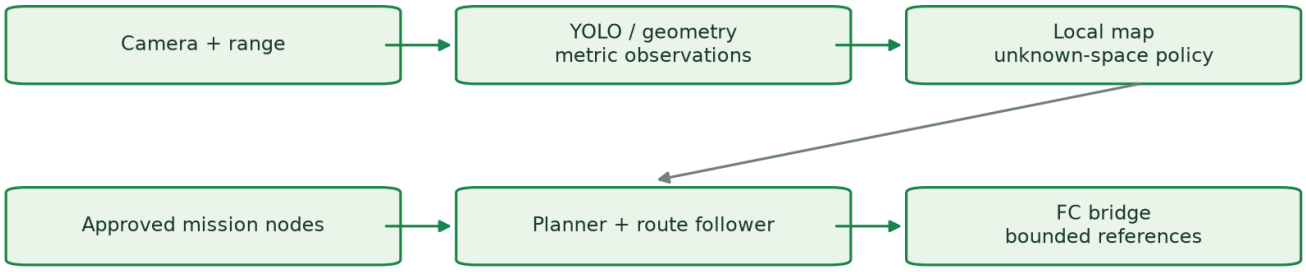
Symbols: SOC: remaining usable-charge fraction; SOC(0): initial fraction; t/τ: elapsed/dummy time (s); I: discharge current (A, positive out); C_usable: usable charge (A s; multiply Ah by 3600); V_ADC/V_pack: divider output/pack voltage (V); R_top/R_bottom: upper/lower divider resistances (ohms).

Measurements: calibrated whole-pack voltage and current, useful capacity, voltage sag and sensor validity. The plots show ideal laws, not measured endurance or an SOC calibration. Marked 120C and four 40 A ESCs do not establish a usable total discharge current.

Historical requested thresholds: warning at 35 percent and motor disarm at 20 percent, without automatic safety landing. A motor cut in flight causes a fall; reconfirm this policy explicitly before any flight phase. Hysteresis, persistence, uncertainty and fault responses remain TBD. Whole-pack software monitoring does not protect individual cells.

Current states: BOOT -> SELF_TEST -> BENCH_READY only. Reset does not arm. Physical motor outputs remain disabled. One verified regulated avionics source; common grounds; no parallel BEC positives; propulsion current stays outside FC copper. Separate authorization is required for motor tests.

2. ROS 2 and YOLO | onboard computer architecture



PROPOSED ARCHITECTURE / The companion requests motion; the RP2350 owns stabilization and motor commands. Real localization remains an integration gate.

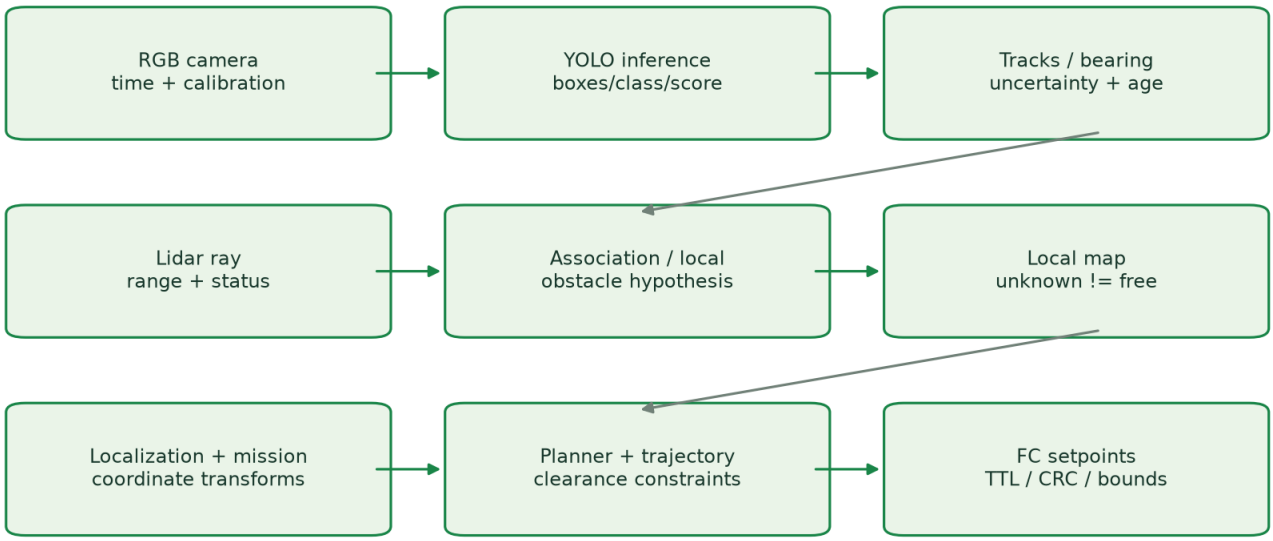
Computer sought: Raspberry Pi 4/5 or NVIDIA Jetson, with RGB camera. Neither is selected. The aircraft FC remains autonomous as a stabilizer without this computer; the computer adds mission planning and obstacle avoidance, not motor-level control.

Layer	Role	Remaining decision
Camera and YOLO	Timestamped images; object hypotheses	Camera calibration, trained panel data, model/runtime
Localization and mapping	Pose; metric obstacle observations; unknown space	Real localization source, scale and uncertainty
Planner and route follower	Replan around obstacles; track approved nodes	Dynamic clearance, expiry and loss policies
FC bridge	Bounded angle/yaw-rate/vertical references	Physical serial acceptance and timing

Portable implementation [P6]: C++ mission, navigator, YOLO and panel-pose nodes, custom messages and F450 description exist. Shared source does not imply identical Pi/Jetson binaries, camera drivers or acceleration backends. Benchmark end-to-end latency, power, mass, cooling and cost before selection.

Mass consequence: adding camera, computer, regulator, cooling and mounts changes mass, centre of mass and inertia. Recompute the supplied 1.038 kg model before gain validation. ROS 2 runs on the companion; RP2350 uses Pico SDK C.

2.1 YOLO | detection, depth and obstacle clearance



PROPOSED ARCHITECTURE / Proposed perception-to-control pipeline. Detection is not depth, localization or a collision-free corridor.

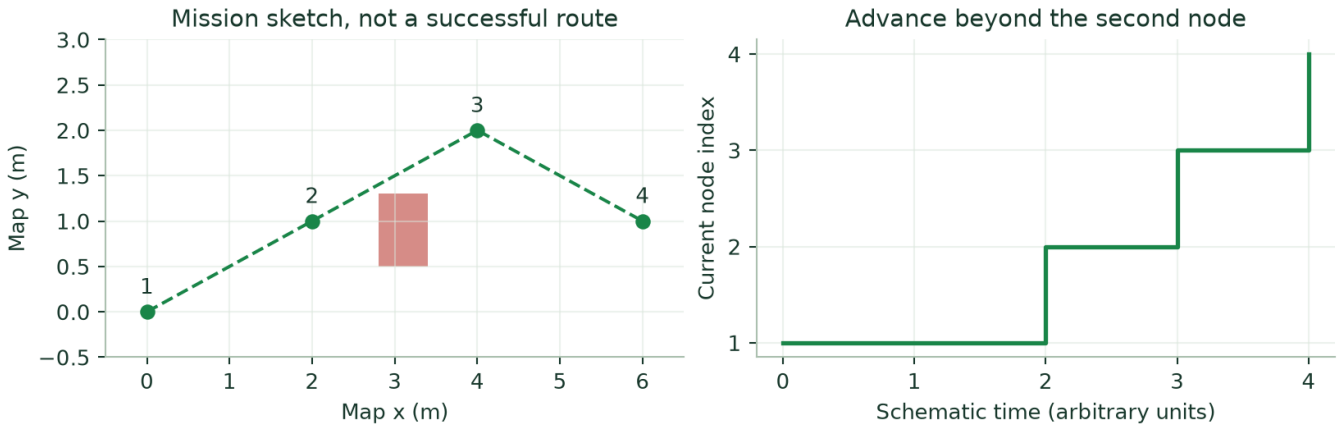
$$\mathbf{d}_c \propto K_c^{-1}[u, v, 1]^T, \quad d_{clear} > v\tau + \frac{v^2}{2a_{brake}} + d_{margin}$$

Symbols: d_c: camera-frame ray direction; K_c: intrinsic camera matrix; u/v inside [u,v,1]: pixel coordinates; d_clear: available clearance (m); v in the stopping law: speed (m/s); τ: total latency (s); a_brake: available braking deceleration (m/s²); d_margin: clearance allowance (m). These local symbols differ from control input u and world velocity vector v.

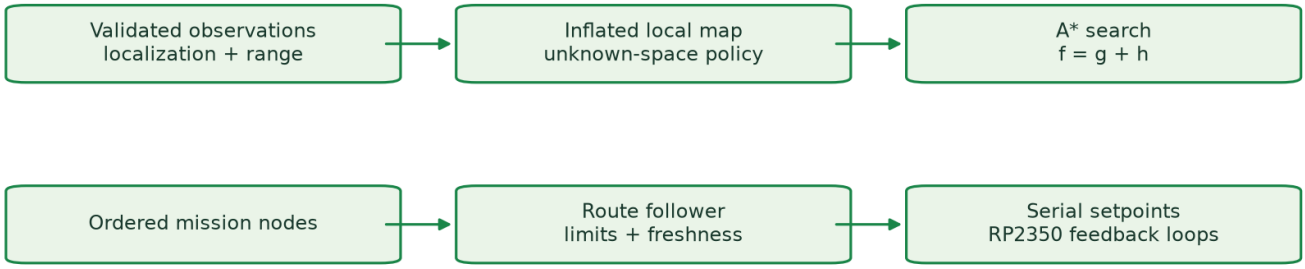
YOLO [R7]: class, box and confidence; not metric free space. Camera calibration, timestamps, tracking and camera/body transforms are required. Associate the single lidar ray only when geometry/time justify it. Unobserved space stays unknown; no detection is not proof of clearance.

Hardware limit: one RGB camera + one fixed ray cannot observe the full collision corridor. Downward lidar aids altitude; forward lidar aids one direction. Placement remains open. Existing known-panel PnP assumes a trained detector and confirmed 1.09 x 1.69 m panel dimensions; it does not recover arbitrary obstacle depth.

2.2 Navigation | approved nodes and online replanning



PLANNED TEST SCHEMATIC / The dashed node sequence is input geometry; it does not claim obstacle avoidance or a collision-free path.



PROPOSED ARCHITECTURE / A bounded C++ A* prototype exists; generalized n-node ROS integration, real localization and physical bridge acceptance remain pending.

$$f(n) = g(n) + h(n), \quad r_{inflate} \geq r_{vehicle} + d_{stop} + k_\sigma \sigma_{position} + d_{margin}$$

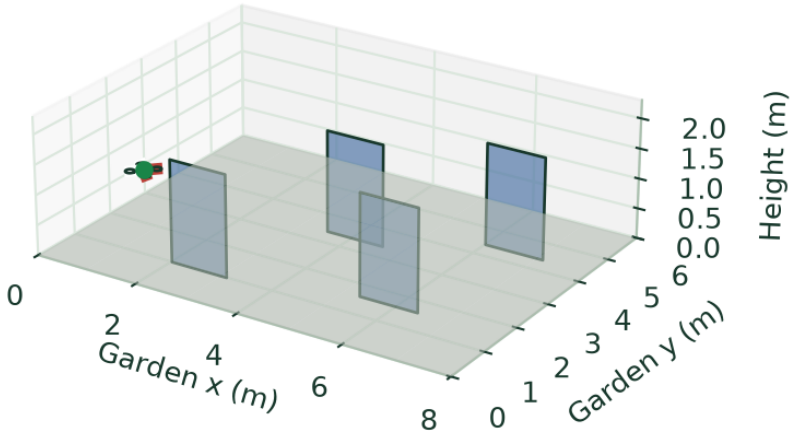
$$\|\hat{\mathbf{p}} - \mathbf{p}_i\| \leq r_{accept}, \quad \|\hat{\mathbf{v}}\| \leq v_{accept} \quad \text{for} \quad t \geq t_{dwell}$$

Symbols: n: search node; f: A* priority; g(n): accumulated path cost, not gravity; h(n): remaining-cost estimate; r_inflate/r_vehicle: map inflation/vehicle radius (m); d_stop/d_margin: stopping distance/margin; kσ: uncertainty multiplier; σ_position: position standard deviation (m); p-hat: estimated position; p_i: node i target; v-hat: estimated velocity; r_accept/v_accept: arrival limits; t/t_dwell: time within limits/required duration.

Target: PC chooses n nodes -> approves mission -> companion localizes, maps, replans and tracks -> FC receives bounded references. Intermediate-node advancement must be tested; final arrival and landing are separate policies.

Existing [P6]: portable A* and waypoint follower; 0.2 m grid, fixed 0.65 m inflation, bounded XY map. These are prototype assumptions, not validated 3D clearance. Current ROS service accepts start/finish only; a cancellable n-node action remains proposed. Proposed arrival: 0.5 m radius and 1 s dwell, speed bound TBD.

3. Gazebo and RViz | model and test environment



ILLUSTRATIVE 3D TEST DESIGN / A proposed scene, not a Gazebo screenshot or successful navigation. Obstacles vary between scenarios and are not given to the planner.

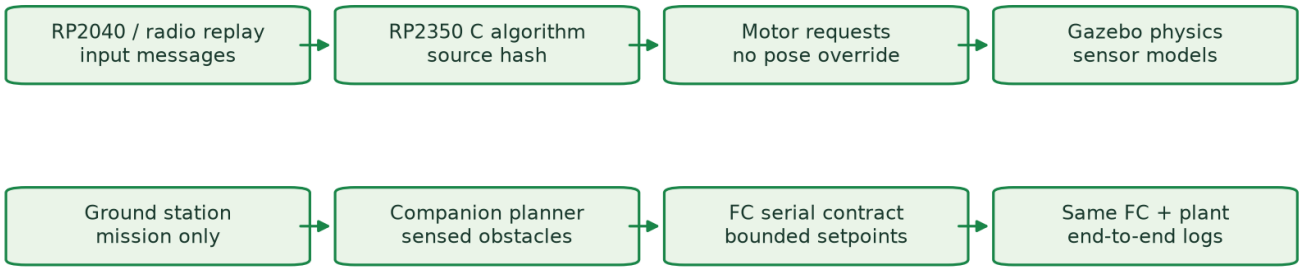
Asset or tool	What it represents	What it cannot prove alone
F450 URDF / RViz	Links, visual geometry, sensor frames, transforms	RViz is a viewer, not a flight physics engine
Gazebo physics model	Mass, inertia, collisions, rotor forces and sensor models	Assumed parameters are not measured hardware
Garden scenario	Panels at varied locations; mission nodes	Illustration is not an accurate USTHB reconstruction
Camera / lidar simulation	Observable images and one lidar ray	Perfect depth/pose must not be granted silently

Model input: m = 1.038 kg; Jxx = Jyy = 0.006421, Jzz = 0.011853 kg m²; nominal F450 diagonal 450 mm; 1400 KV motors/9047 props/40 A ESCs. Thrust curve, motor lag, exact battery/companion mass and drag remain unmeasured.

Construction: visual/collision geometry -> measured mass/inertia -> motor dynamics -> sensor noise/delay -> actual FC algorithm -> test command sources. URDF supports visualization; the Gazebo model/plugins must independently implement forces and sensors.

Figure: illustrative 3D garden and F450 geometry generated for this report; not a Gazebo screenshot, rendered mission or reconstruction. Panel locations are test parameters, hidden from the navigation input.

3.1 Validation | prove the actual FC drives the plant



PROPOSED ARCHITECTURE / A test harness may stimulate inputs and record outputs; it may not secretly move the aircraft to pass.

Campaign	Stimulus	Required evidence
Manual: 5 scenarios	Simulated pilot messages only	Actual FC estimates/PID/motor requests; takeoff, motion, landing
Autonomous: 10 environments x 5 seeds	Approved mission nodes only	Companion perception/planning -> FC setpoints -> plant
Fault injection	Noise, stale data, link loss, invalid ranges	Bounded response; explicit verdict; retain failures

No hidden help: no pose teleportation, injected route-following forces or script that bypasses the FC. Record source/configuration hashes, command timestamps, sensor inputs, controller states, motor requests and plant truth. Truth is available for scoring, not secretly fed to autonomy. Host C execution is not MCU instruction emulation or timing proof.

Proposed targets: attitude RMS <=2°, height RMS <=0.20 m, endpoint <=0.5 m; zero collisions in the declared campaign. RMS means root mean square of tracking error. These are targets, not achieved results. No new Gazebo campaign was run for this edition. Video at 10-15 fps aids review; full-rate logs establish timing.

Timing evidence [P3]: historical ESP32 acquisition handler maxima 814-822 µs; earlier failed trials 1381/1249 µs retained. This is neither full FC timing nor RP2350 profiling. The inspected Pico scaffold remains 500 Hz with hardware stubs; 1 kHz requires fresh integration/load tests.

Progression rule: power -> calibration/frame -> estimation/timing -> PID/mixer -> manual -> navigation. Failure invalidates affected downstream acceptance. No physical motor authorization is granted by this report.

4. Resources | hardware and components

Subsystem	Intended materials	Selection status
Airframe and propulsion	F450 red/white; 4 x 1400 KV motors; 4 x 9047 props; 4 x 40 A ESC	Exact motor/ESC models and dynamics to verify
Power	3S 4200 mAh LiPo, marked 120C; verified avionics regulator; voltage/current sensing	Actual capacity, sag, mass and current limits to measure
Aircraft control	RP2350 board, MPU6050, BMP280, GY-271 magnetometer, TOFSense F2 mini	Exact boards/compass silicon and drivers to confirm
Handheld	RP2040 board; 2 sticks; 2 triggers; buttons; display; power; 74HC4051	Display/power/mux validation open
Radio totals	2 LoRa; 4 CC1101; 6 matched antennas	Exact modules/band/antennas open; CC1101 reserved
Autonomy option	Pi 4/5 OR Jetson, RGB camera, storage, regulator, cooling, mounts	Being sought; not purchased/selected
Integration and measurement	PCB/headers/wire; multimeter; scale; level/fixtures; reference instruments	Full 37-entry inventory in CSV

$$C_{base} = \sum_{i} n_i c_i + C_{shipping} + C_{fabrication}$$

Symbols: C_base: base project cost; i: inventory item; n_i: quantity; c_i: unit price; C_shipping/C_fabrication: shipping/fabrication costs. All cost terms use one stated currency; no numeric total exists until quotations are entered.

Budget: no fabricated prices. Dated quotes, owned/to-buy status, shipping and measured mass belong in bill-of-materials.csv. Shared tools and optional companion are costed separately.

Mass update: replace prototype FC mass; add radios, antennas, camera, computer, regulator and mounts. Recompute centre of mass and inertia before gain validation.

4.1 Resources | software and professional method

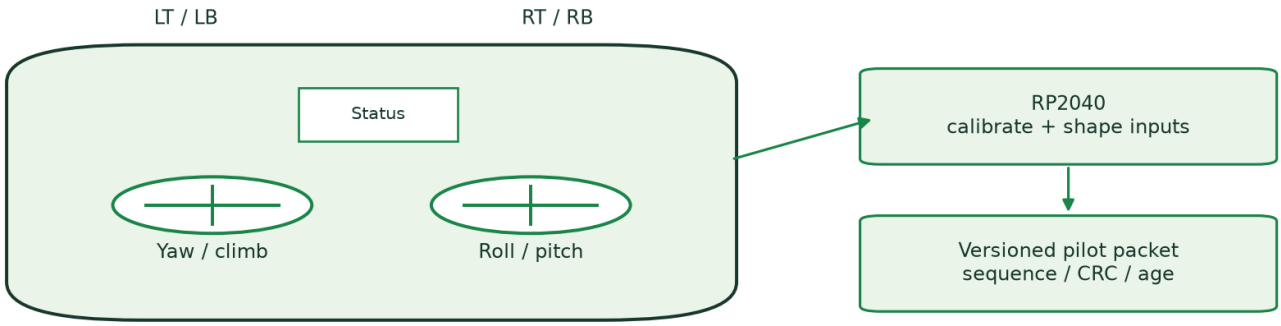
Software	Project use	Engineering output
Pico SDK / C toolchain / CMake	RP2350 FC; RP2040 handheld	Reproducible firmware build and pinned versions
ROS 2 / C++	Companion mission, navigation, bridge	Packages, interface contracts, bounded message age
YOLO inference runtime	Camera detections	Model hash, training scope, measured latency
Gazebo	Dynamic plant, motor and sensor simulation	Repeatable scenarios, physics configuration, logs
RViz	F450 URDF, frames, markers and sensor views	Visual checks; no implied physics simulation
Python / NumPy / Matplotlib	Calibration, uncertainty and plots	Repeatable numerical analyses and retained adverse results
Git / GitHub	Version control and review	Descriptive commits, source hashes, no force-push
KiCad	User-designed electronics	Module dimensions, clearances and connectivity checks

Methods reused: sensor/radio libraries, ROS middleware and inference frameworks avoid unnecessary reinvention. Record exact versions, licences and configuration. Software availability does not prove its hardware interface or flight behaviour.

Professional contribution: formulate requirements; derive dynamics; design calibration experiments; interpret uncertainty and failures; integrate perception, control and testing. Complete the personal reflection with actual actions/results, not invented experience. Attribute libraries and AI assistance under university policy.

Acceptance dossier: 50 requirements, 13 verification cases, 37-entry materials inventory, original evidence provenance and source fingerprints. Costs remain TBD until dated quotations are recorded. An attractive video complements, but does not replace, reproducible data.

5. Handheld | RP2040 pilot interface



PROPOSED USER INTERFACE / Two centred sticks, triggers/buttons and status display. Exact mechanical design and gesture mapping remain to select.

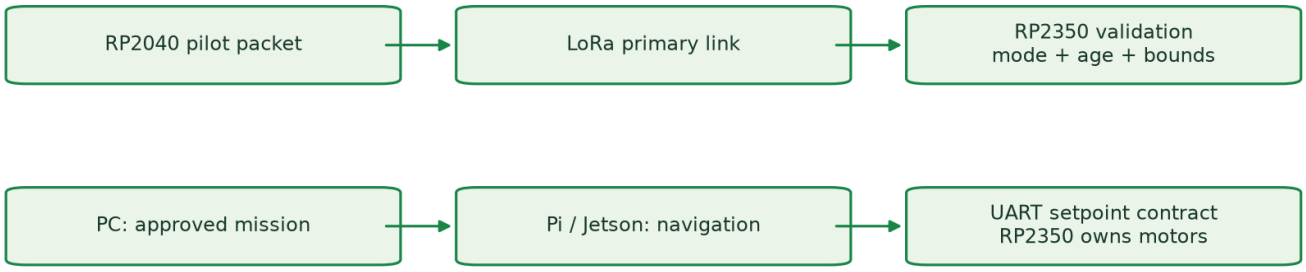
Input	Intended behaviour	Validation
Right stick	Pitch / roll angle reference	Neutral/dead zone; axis sign; gentle limits
Left horizontal	Yaw-rate reference	Correct direction and rate bound
Left vertical	Vertical-speed reference	Centre height retention through FC vertical law
RT / LT	Bounded increase / decrease	Preserve PID headroom; no direct motor bypass
LB / RB and extra buttons	Mode/request functions	Final RP2040 gestures remain to define
Display	Mode, battery and link status	Must not block input or command handling

Acquisition: six analogue channels (four stick axes + two triggers) require an acquisition arrangement; the existing 74HC4051 mux needs channel settling and calibration checks. Power and display models remain to select.

Command path: centre/scale -> dead zone -> gentle response -> limits -> packet. FC validates freshness and mode before using references. Sticks request motion; the FC retains its own stabilization and motor authority. Mode changes require deliberate validated gestures, not connection alone.

Independence: no Pi/Jetson is needed for stable pilot mode. The companion may be absent at boot. Physical motor arming remains disabled during the current bench stage.

5.1 Links | LoRa, CC1101 and companion contract



PROPOSED ARCHITECTURE / Two separate authorities: pilot references and selected-mode companion references. CC1101 secondary links remain reserved.

$$T_{sym} = \frac{2^{SF}}{BW}, \quad T_{age} = t_{now} - t_{sample} < T_{valid}$$

Symbols: Tsym: LoRa symbol duration (s); SF: spreading factor; BW: bandwidth (Hz); t_now/t_sample: receiver time/sample time on a consistent clock; T_age: command age; T_valid: allowed age. Symbol duration is not complete packet latency.

Pilot link: 1 LoRa module per side. Versioned fields: sequence, normalized controls, buttons, CRC and validity. Measure full airtime, queueing, loss bursts, interference, reconnect and supply integrity. The legacy 22-byte frame needs an explicit versioned extension for new fields. Radio cadence is slower than the local 1 kHz target loop.

Secondary links: 2 CC1101 modules per side, reserved for later data. Their exact modules/roles and coexistence remain unvalidated; they are not automatic safety redundancy. Select matched band/antennas and verify applicable RF constraints.

Companion link: proposed framed 3.3 V UART with length, CRC, sequence, units, mission identity, timestamp and expiry. Proposed heartbeat 10 Hz/expiry 300 ms require validation. Invalid frames do not refresh validity. ROS QoS uses bounded histories and freshness; reliable delivery does not guarantee a deadline.

Authority contract: PC uploads approved nodes; companion navigates; FC accepts bounded references only in the selected mode and owns motor commands. Plugging in the companion does not select autonomous mode. Loss policies and final-arrival actions remain to approve; manual FC must remain usable without it.

References and reproducible evidence

Public engineering repository
https://github.com/BenMohStem/IntelQCopter/tree/codex/usthb-university-report
Source, versions, report, measurements and scoped commit history.

Institution: USTHB; Faculté de Génie Électrique; Département Automatique; Robotique et Systèmes Intelligents (RSI), fourth-year engineering. Official programme listing: https://fge.usthb.dz/pages/Orientation-2026-2027 . Published RSI curriculum (2025 update), PDF p. 11: Projet Personnel Professionnel, RSI7.8, semester 7, 2 credits, coefficient 1. Source: https://fge.usthb.dz/storage/disk/Robotique-et-systemes-intelligents-MAJ-2025.pdf . The teacher's assessment rubric remains to obtain.

Primary technical sources: [R3] TDK MPU6050 register map: https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Register-Map1.pdf . [R4] Bosch BMP280: https://www.bosch-sensortec.com/products/environmental-sensors/pressure-sensors/bmp280/ . [R5] Nooploop TOFSense: https://support.nooploop.com/tofsense/ . [R7] YOLO detection: https://docs.ultralytics.com/tasks/detect/ .

[R8] MIT OCW feedback control: https://ocw.mit.edu/courses/16-30-feedback-control-systems-fall-2010/ . [R9] ROS frames: https://www.ros.org/reps/rep-0105.html . [R10] RP2350: https://www.raspberrypi.com/products/rp2350/ . [R11] CC1101: https://www.ti.com/product/CC1101 . [R12] Semtech LoRa: https://www.semtech.com/design-support/faq/faq-lora . [R13] ROS 2 concepts: https://docs.ros.org/en/jazzy/Concepts.html . These support methods, not acceptance of this aircraft.

Project evidence: [P1] secured-mount six-pose calibration; [P2] gyro restart validation; [P3] FIFO/frame/timing; [P4] reserved recording replay and adverse synthetic models; [P5] input guard checks. Raw evidence is distinguished from derived summaries. [P6] inspected Pico/ROS source commit bce665ad97adbfcb282f02970a1db599e67e6479; source-inventory.json records exact files. No fresh RP2350 flight, hardware-radio acceptance or complete autonomous mission is claimed.

Reproduce: report/report.md is editable; tools/build_report.py rebuilds the PDF/HTML; report/ data files and manifests retain requirements, inventory, assumptions and hashes. Original private recordings require a sanitized examiner evidence package. Cover = AI F450 concept, not a hardware photograph. Mathematical schematics and proposed test environments are labelled. Report edition: 27 October 2026; institutional/source checks: 28 October 2026.